Computer Forensics Case Studies

Case Study 2 Reconstructing Chrome Web History

Student Assessment Brief

HUSSEINI SULEIMAN

2025/FWSD/11478

fwsd2511478@students.icdfa.edu.ng

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SBTDF204_CaseStudy2_**2025/FWSD/11478**

Assessment detail	Information
Course	SBT-DF204 Computer Forensics Case Studies
Assessment	Case Study 2 Individual browser history investigation
Case title	Reconstructing Suspected Drug Trafficking From Chrome Web History
Submission format	Written report with SQL evidence, screenshots, timeline, and evidence log
Deadline and maximum score	As published in E-Campus

This individual case study asks you to examine a Chrome History SQLite database and reconstruct an evidence-based timeline of suspected drug trafficking activity. You will identify meaningful browser records, convert timestamps, examine navigation transitions, and connect observations across postings, communications, payment

references, and downloads. Your report must distinguish what the browser history directly records from what you infer about the user or alleged activity.

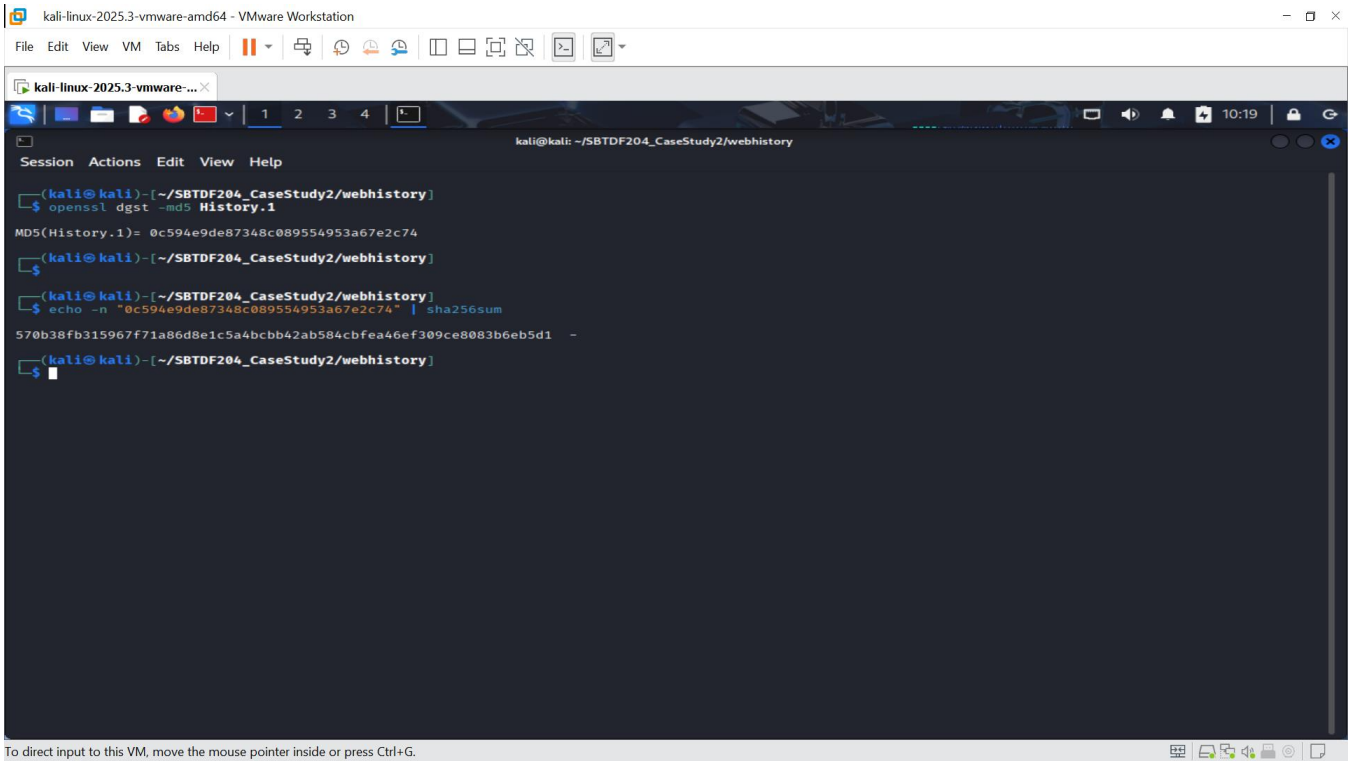
Case scenario

Investigators received a Google Chrome History file from a device associated with a suspect. The case hypothesis states that the browser may have been used to advertise controlled drugs, communicate with an interested buyer, exchange payment information, verify a cryptocurrency transaction, and retain proof of payment. Your task is to assess the available history evidence and construct a defensible timeline. The case requires forensic interpretation of a supplied database, not interaction with live websites, accounts, messages, or wallets.

Case materials

The supplied case file is a Chrome History SQLite database named **History**. The teaching source shown in the case materials is: added in study material section

Create a working copy before opening the database. Record the source, original file name, file size, and SHA-256 value. Use a SQLite examination tool such as DB Browser for SQLite or sqlite3. Preserve screenshots and SQL output that support your findings.



What you must submit

Item	Required output	What to provide
1	Forensic working copy	Acquire the supplied Chrome History file. Record its source: Chrome History SQLite database original file name: **History** byte size: 196608 SHA-256 value: 570b38fb315967f71a86d8e1c5a4bcbb42ab584cbfea46ef309ce8083b6eb5d1 The location of your read-only original: Kali/Downloads/ SBTDF204_CaseStudy2/history.

Item	Required output	What to provide
2	Database examination	Identify the relevant SQLite tables and their fields. Your report must use the urls, visits, and downloads evidence that applies to your findings.
3	Reconstructed timeline	Build a time-ordered account of the browser activity. Cite visit IDs, URL IDs, timestamps, transition values, titles, URLs, and download record IDs where relevant.
4	Student report	Submit one report in PDF or DOCX format, with SQL output or screenshots, a completed evidence log, and a timeline appendix.

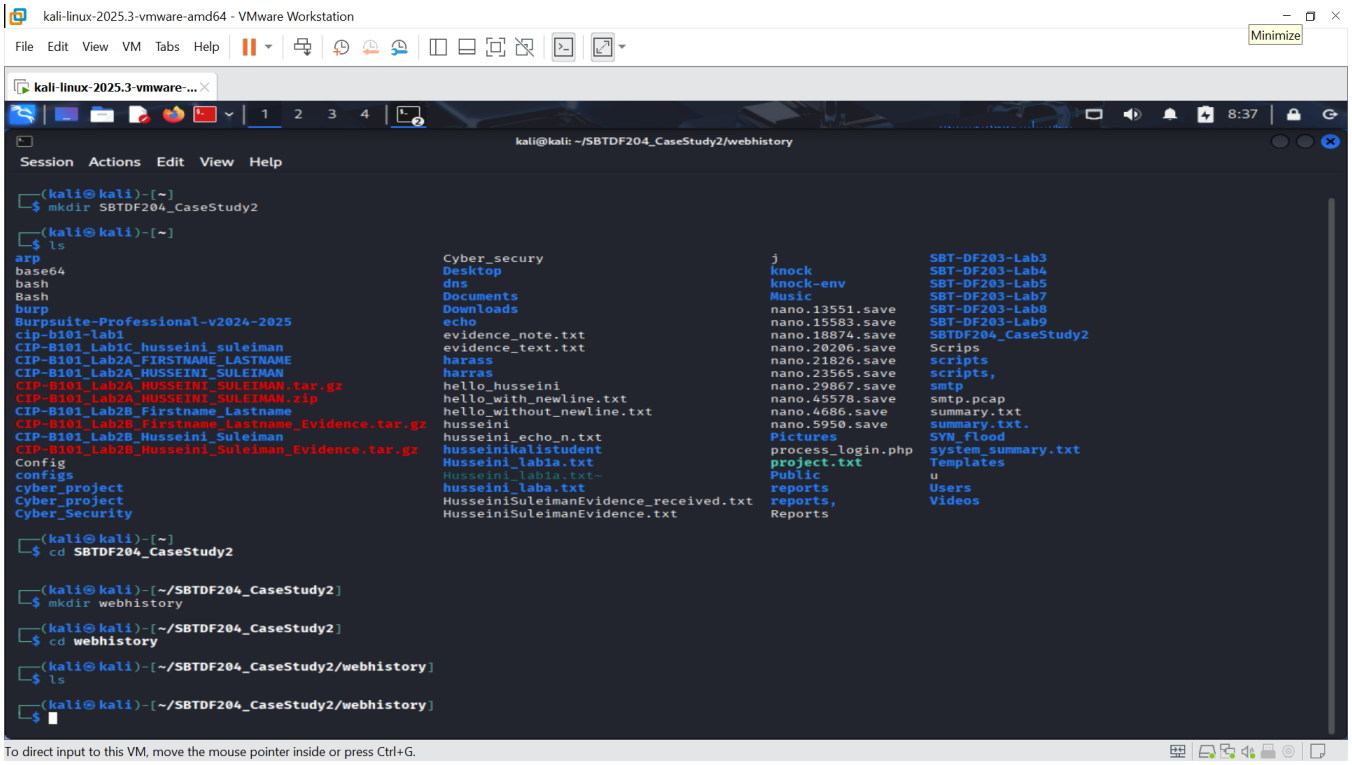
Investigation task

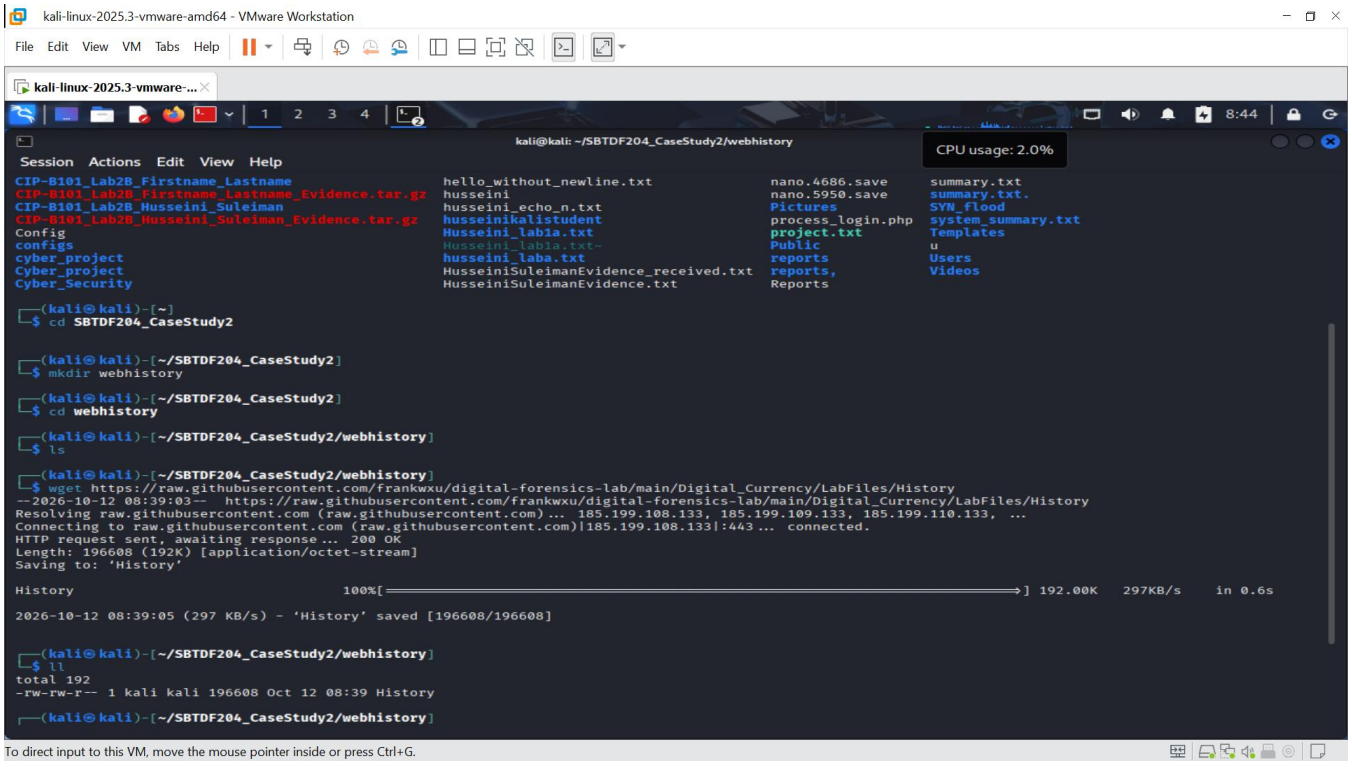
Examine the History database methodically. Base all findings on record IDs, table fields, URLs, titles, timestamps, transitions, and related download artefacts. A URL or title can indicate browser activity, but it may not prove the full content of a page, the person operating the device, or the real-world completion of an alleged transaction. Use the questions below to structure your report.

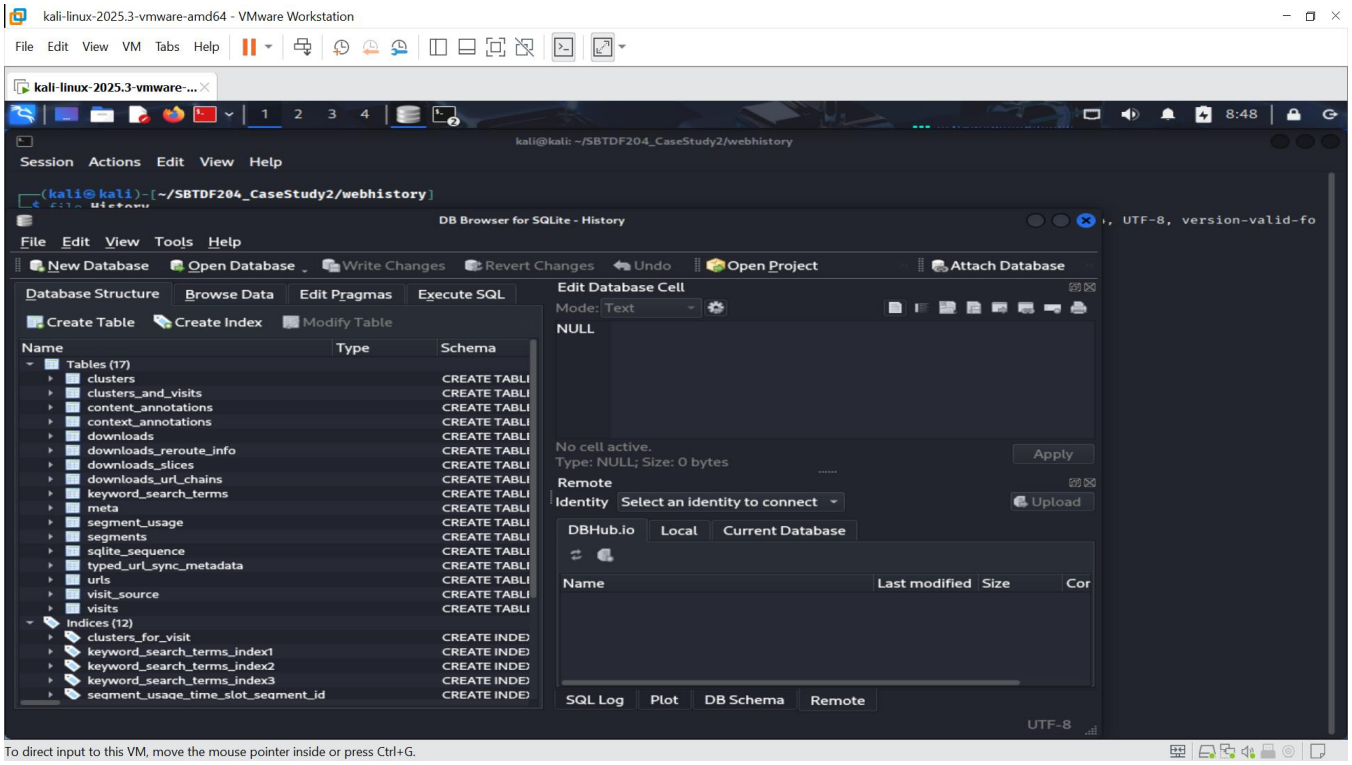
No	Investigation question	Minimum evidence to discuss
1	What evidence file did you acquire and how did you preserve it?	Report the original file name, source, file size, SHA-256 value, working-copy method, and any file-access precautions.

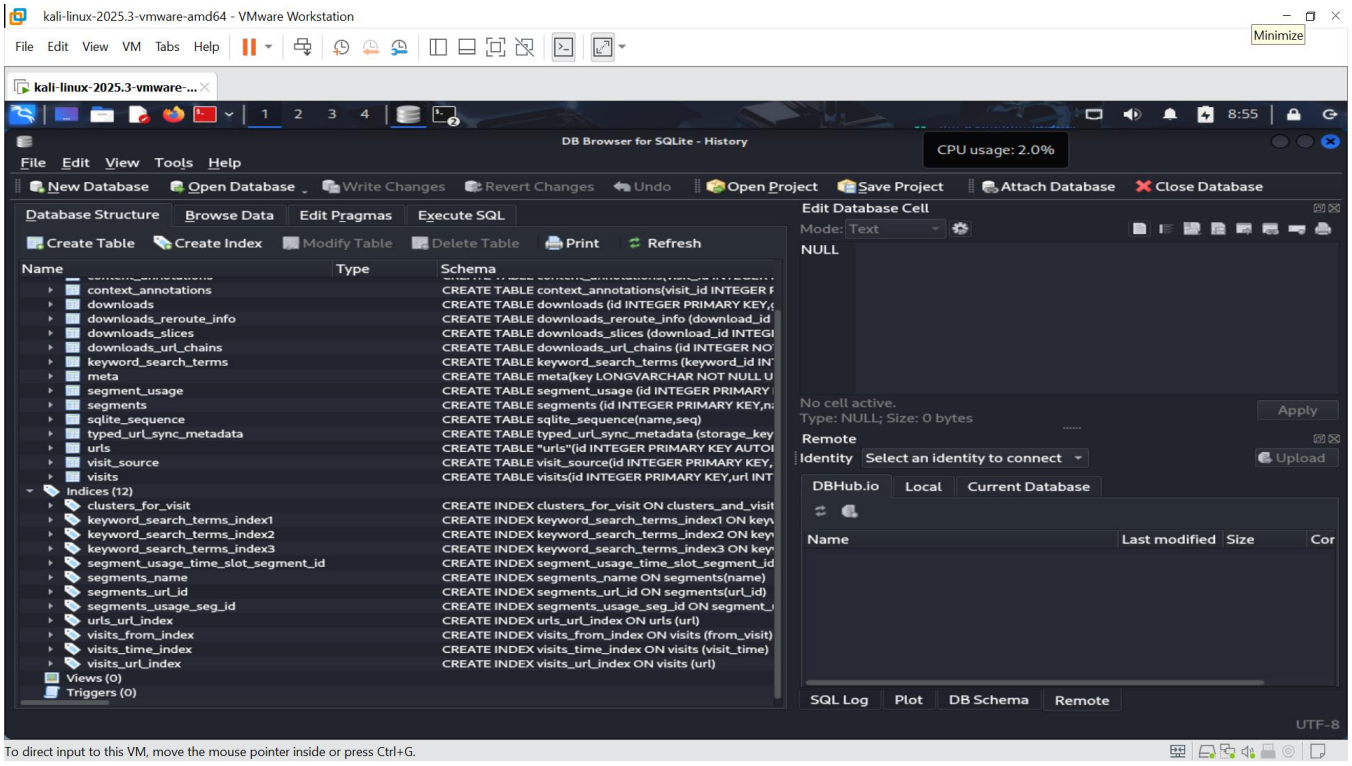
No	Investigation question	Minimum evidence to discuss
2	What browser-history tables and fields are relevant?	Identify the tables you examined and explain the role of the urls, visits, downloads, and downloads_url_chains data used in your case.
3	When did the recorded web activity occur?	Convert Chrome WebKit timestamps to UTC and provide the query or method. State any local-time conversion separately.
4	How did the browser reach the material URLs?	Use transition values, transition qualifiers, and from_visit links to distinguish typed or omnibox activity, links, form submissions, redirects, and navigation chains.
5	What web-history evidence relates to the suspected sale or posting?	Identify the records that support or challenge an advertisement, posting workflow, management page, or related interaction. Cite the exact artefacts.
6	What evidence relates to communications and payment activity?	Identify relevant email, image-hosting, blockchain-explorer, exchange, transaction, wallet, or download artefacts. Explain each link without assuming content that the record does not show.
7	What is the reconstructed timeline?	Create a chronological table that separates direct observations from your interpretation of the activity stage.

No	Investigation question	Minimum evidence to discuss
8	What conclusion can you defend?	Give a cautious conclusion with confidence level and at least three limitations or alternative explanations, including the difference between browser-profile activity and human identity.


kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Minimize
kali-linux-2025.3-vmware-...
8:37
kali@kali: ~/SBTDF204_CaseStudy2/webhistory
Session Actions Edit View Help
(kali㉿kali)-[~]
$ mkdir SBTDF204_CaseStudy2
$ ls
arp
base64
bash
Bash
burp
Burpsuite-Professional-v2024-2025
cip-b101-lab1
CIP-B101_Lab1C_husseini_suleiman
CIP-B101_Lab2A_FIRSTNAME_LASTNAME
CIP-B101_Lab2A_HUSSEINI_SULEIMAN
CIP-B101_Lab2A_HUSSEINI_SULEIMAN.tar.gz
CIP-B101_Lab2A_HUSSEINI_SULEIMAN.zip
CIP-B101_Lab2B_Firstname_Lastname
CIP-B101_Lab2B_Firstname_Lastname_Evidence.tar.gz
CIP-B101_Lab2B_Husseini_Suleiman
CIP-B101_Lab2B_Husseini_Suleiman_Evidence.tar.gz
Config
configs
cyber_project
Cyber_project
Cyber_Security
Cyber_secury
Desktop
dns
Documents
Downloads
echo
evidence_note.txt
evidence_text.txt
harass
harras
hello_husseini
hello_with_newline.txt
hello_without_newline.txt
husseini
husseini_echo_n.txt
husseinikalistudent
Husseini_lab1a.txt
Husseini_lab1a.txt~
husseini_laba.txt
HusseiniSuleimanEvidence_received.txt
HusseiniSuleimanEvidence.txt
j
knock
knock-env
Music
nano.13551.save
nano.15583.save
nano.18874.save
nano.20206.save
nano.21826.save
nano.23565.save
nano.29867.save
nano.45578.save
nano.4686.save
nano.5950.save
Pictures
process_login.php
project.txt
Public
reports
reports,
Reports
SBT-DF203-Lab3
SBT-DF203-Lab4
SBT-DF203-Lab5
SBT-DF203-Lab7
SBT-DF203-Lab8
SBT-DF203-Lab9
SBTDF204_CaseStudy2
Scrips
scripts
scripts,
smtp
smtp.pcap
summary.txt
summary.txt.
SYN_flood
system_summary.txt
Templates
u
Users
Videos
$ cd SBTDF204_CaseStudy2
(kali㉿kali)-[~/SBTDF204_CaseStudy2]
$ mkdir webhistory
(kali㉿kali)-[~/SBTDF204_CaseStudy2]
$ cd webhistory
(kali㉿kali)-[~/SBTDF204_CaseStudy2/webhistory]
$ ls
(kali㉿kali)-[~/SBTDF204_CaseStudy2/webhistory]
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
8:44
kali@kali: ~/SBTDF204_CaseStudy2/webhistory
CPU usage: 2.0%
Session Actions Edit View Help
CIP-B101_Lab2B_Firstname_Lastname
CIP-B101_Lab2B_Firstname_Lastname_Evidence.tar.gz
CIP-B101_Lab2B_Husseini_Suleiman
CIP-B101_Lab2B_Husseini_Suleiman_Evidence.tar.gz
Config
configs
cyber_project
Cyber_project
Cyber_Security
hello_without_newline.txt
husseini
husseini_echo_n.txt
husseinikalistudent
Husseini_lab1a.txt
Husseini_lab1a.txt~
husseini_laba.txt
HusseiniSuleimanEvidence_received.txt
HusseiniSuleimanEvidence.txt
nano.4686.save
nano.5950.save
Pictures
process_login.php
project.txt
Public
reports
reports,
Reports
summary.txt
summary.txt.
SYN_flood
system_summary.txt
Templates
u
Users
Videos
$ cd SBTDF204_CaseStudy2
$ mkdir webhistory
$ cd webhistory
$ ls
$ wget https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Digital_Currency/LabFiles/History
--2026-10-12 08:39:03-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Digital_Currency/LabFiles/History
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 196608 (192K) [application/octet-stream]
Saving to: 'History'
History 100%[] 192.00K 297KB/s in 0.6s
2026-10-12 08:39:05 (297 KB/s) - 'History' saved [196608/196608]
$ ll
total 192
-rw-rw-r-- 1 kali kali 196608 Oct 12 08:39 History
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.





Chrome History analysis starting points

Chrome stores WebKit timestamps as microseconds since 1 January 1601 UTC. Use UTC for the main investigation timeline. If you also

display local time, state the time zone and conversion. Inspect the schema first because Chrome versions can differ. The following SQL provides suitable starting points, not a substitute for examining the returned records.

Purpose	Starting SQL

List the available tables	`SELECT name`
	`FROM sqlite_master`
	`WHERE type = 'table'`
	`ORDER BY name;`

Inspect a table schema	`PRAGMA table_info(urls);`
	`PRAGMA table_info(visits);`
	`PRAGMA table_info(downloads);`

Build a visit timeline	`SELECT v.id AS visit_id, u.id AS url_id, u.url, u.title,`
	
	`datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS visit_utc,`
	`v.from_visit, v.transition, v.visit_duration/1000.0 AS duration_ms`
	`FROM visits AS v JOIN urls AS u ON v.url = u.id`
	`ORDER BY v.visit_time;`

Review downloaded artefacts	`SELECT *`
	`FROM downloads`
	`ORDER BY start_time;`

Transition interpretation

Use the transition field together with from_visit. To determine the basic transition type, calculate transition AND 0xFF. The case materials demonstrate base values such as 0 for a link, 5 for omnibox-generated activity, and 7 for a form submission. Qualifier bits can indicate

navigation-chain start, server redirect, or refresh behaviour. State the exact value you found and explain the interpretation you used. Do not rely on the transition value alone when the URLs, titles, and linked visits provide a different context.

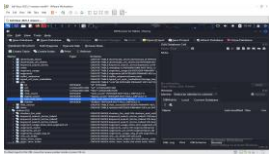
Required report structure

1. Case reference and student identification. Include your full name, registration number, course code, and case title.
2. Executive conclusion. State the supported conclusion in a short, cautious paragraph. Do not treat browser-profile activity as conclusive proof of a named person.
3. Evidence acquisition and integrity. Record the History source, file name, size, hash, working-copy method, and tools used.
4. Database method. List the tables reviewed and explain your SQL, timestamp conversion, joins, and transition interpretation.
5. Findings. Answer each investigation question with record IDs, URLs or titles, timestamps, transition values, SQL output, and evidence-log IDs.
6. Timeline. Present the relevant browser sequence in UTC and distinguish the observed artefact from your interpretation.
7. Attribution assessment. State a confidence level and explain limitations, including shared-device, browser-profile, redirect, cached-content, and missing-content considerations where relevant.
8. Appendices. Include labelled screenshots, selected SQL output, the completed timeline table, and the completed evidence log.

Timeline template

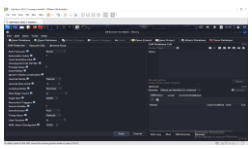
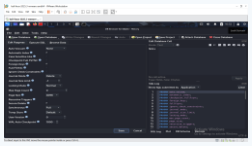
Use a completed version of this table in your report or evidence appendix. Add rows for all material events.

Stage	UTC timestamp	Visit or URL ID	Artifact	Observed fact	Interpretation and limit

Stage	UTC timestamp	Visit or URL ID	Artifact	Observed fact	Interpretation and limit
		LONG VARCHAR			

Evidence log template

Use an evidence log that lets an assessor locate the exact database record, query result, or screenshot supporting each finding.

Evidence ID	Source record	Finding	Why it matters	SQL or screenshot reference
E01				
E02				
E03				
E04				
E05				

Evidence ID	Source record	Finding	Why it matters	SQL or screenshot reference
E06				

Submission rules

- Work individually. Your SQL analysis, report wording, screenshots, timeline, and evidence log must be your own.
- Submit the report as SBTDF204_CaseStudy2_RegistrationNumber.pdf or .docx. If you submit separate supporting material, use SBTDF204_CaseStudy2_Evidence_RegistrationNumber.zip.
- Do not upload the original History database unless your lecturer specifically requests it. Submit only the report and the required supporting evidence.
- Use this database only as authorised course material. Do not visit or interact with live postings, people, email accounts, exchanges, wallets, transaction IDs, or external platforms referenced in the history.
- Redact unrelated personal data that may appear in screenshots while retaining enough record detail for the examiner to verify your finding.

Marking guide

The E-Campus maximum score applies to this case study. The assessor will apply the following percentage weighting to that maximum score.

Criterion	What strong work demonstrates	Weight

Criterion	What strong work demonstrates	Weight
Evidence acquisition and database structure	Records integrity and accurately explains the relevant Chrome History tables and fields.	15%
SQL examination and time conversion	Uses defensible queries, joins records accurately, and converts WebKit timestamps correctly.	20%
Browser action and navigation analysis	Interprets transitions, from_visit links, redirects, omnibox activity, and form submissions with evidence.	20%
Event reconstruction and correlation	Builds a clear timeline and correlates records that relate to the suspected posting, communications, payment, and download activity.	20%
Attribution assessment and limitations	Separates observation from inference, explains alternative explanations, and avoids claiming more than the history supports.	15%
Report quality and evidence presentation	Writes clearly, labels SQL output and screenshots, cites record IDs, and submits a usable evidence log.	10%

Final check before submission

- Every major conclusion cites the relevant table, record ID, timestamp, SQL result, and screenshot or appendix reference.
- Your timeline uses UTC consistently and records any local-time conversion separately.
- Your report explains how browser transitions and from_visit links support the navigation sequence.

- Your conclusion clearly separates observed browser artefacts from inferences about a person or alleged conduct.